

LexiQuery Architecture Specification - Chapter 1: Introduction

LexiQuery is a privacy-first, local-first RAG (Retrieval-Augmented Generation) document assistant. It enables natural language Q&A over complex PDFs using LangChain, Ollama (Llama 3 8B), and ChromaDB. No document content leaves the user local machine during indexing, embedding, or inference.

LexiQuery Architecture Specification - Chapter 2: Technical Design

1. Document Ingestion: PyMuPDF extracts text page by page.
2. Text Chunking: RecursiveCharacterTextSplitter divides text into chunks of 1000 characters with 200 overlap.
3. Vector Storage: Nomic Embed Text generates 768-dim embeddings stored in ChromaDB vector collection.
4. Prompt Pipeline: Hallucination-resistant grounded prompt enforces strict context fidelity.